MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ECL, QUAD TRANSLATOR, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, ECL, logic gating microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Quad translator, TTL to ECL
02	Quad translator, ECL to TTL

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16 pin, 1/4" X 7/8"), dual-in-line package
F	F-5 (16 pin, 1/4" X 3/8"), flat-package

1.3 Absolute maximum ratings.

Supply voltage range	0 V dc to -7.0 V dc for V_{EE} 0 V dc to +7.0 V dc for V_{CC}
Input voltage range	
Device type 01	+5.0 V dc to GND
Device type 02	GND to V_{EE}
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) per gate 1/	
Device type 01	145 mW
Device type 02	130 mW
Lead temperature (soldering, 10 seconds)	+260°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E and F	See MIL-M-38510, appendix C.
Case 2	0.08°C/W 2/
Junction temperature (T_J)	+165°C 3/
Maximum output current	
Device type 01	-50 mA
Device type 02	+24 mA

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	-5.46 V minimum to +5.5 V maximum
Minimum high-level input voltage (V_{IH})	-1.105 V at T_A = +25°C (ECL), 2.0 V (TTL)
(at 500 linear ft/min)	-1.000 V at T_A = +125°C (ECL), 2.0 V (TTL)
(ft/min)	-1.255 V at T_A = -55°C (ECL), 2.0 V (TTL)
Maximum low-level input voltage (V_{IL})	-1.475 V at T_A = +25°C (ECL), 0.8 V (TTL)
(at 500 linear ft/min)	-1.400 V at T_A = +125°C (ECL), 0.8 V (TTL)
	-1.510 V at T_A = -55°C (ECL), 0.8 V (TTL)
Normalized fanout (each output)	10 4/
Case operating temperature range (T_C) (at 500 linear ft/min)	-55°C to +125°C
Case operating temperature range (T_C) (at still air)	
Device type 01	-55°C to +105°C (case E)
	-55°C to +115°C (case F)
Device type 02	-55°C to +115°C (case E)
	-55°C to +120°C (case F)

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be specified on figure 2.

4/ Device will fanout in both high and low levels to the specified number of data inputs on the same device type as that being tested.

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3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

TABLE I. Electrical performance characteristics.

(Limits are valid provided circuit is in a test socket and transverse air flow of 500 linear ft/min is maintained.)

Test	Symbol	Conditions 1/ 2/ $-55^{\circ}C \leq T_C \leq +125^{\circ}C$	T_C	V_{IH1}	V_{IL1}	Device type	Limits Min	Limits Max	Units
High level output voltage	V_{OH}	V_{EE} = -5.2 V V_{CC} = +5.0 V Load = 100Ω to -2 V	+25°C +125°C -55°C	+2.4 V +2.4 V +2.4 V	+0.4 V +0.4 V +0.4 V	01	-0.930 -0.825 -1.080	-0.780 -0.630 -0.880	V
		V_{EE} = -5.2 V V_{CC} = +5.0 V Load = -2 mA	+25°C +125°C -55°C	-0.780 -0.630 -0.880	-1.850 V -1.820 V -1.920 V	02	2.5 2.5 2.5	--- --- ---	V
			T_C	V_{IH1}	V_{IL1}				
Low level output voltage	V_{OL}	V_{EE} = -5.2 V V_{CC} = +5.0 V Load = 100Ω to -2 V	+25°C +125°C -55°C	+2.4 V +2.4 V +2.4 V	+0.4 V +0.4 V +0.4 V	01	-1.850 -1.820 -1.920	-1.620 -1.545 -1.655	V
		V_{EE} = -5.2 V V_{CC} = +5.0 V Load = +12 mA	+25°C +125°C -55°C	-0.780 -0.630 -1.880	-1.850 V -1.820 V -1.920 V	02	--- --- ---	0.5 0.5 0.5	V
			T_C	V_{ITH}	V_{ITL}				
High level threshold output voltage	V_{OTH}	V_{EE} = -5.2 V V_{CC} = +5.0 V Load = 100Ω to -2 V	+25°C +125°C -55°C	+1.8 +1.8 +2.0	+1.0 +0.80 +1.1	01	-0.950 -0.845 -1.100	--- --- ---	V
		V_{EE} = -5.2 V V_{CC} = +5.0 V Load = -2 mA	+25°C +125°C -55°C	-1.105 V -1.000 V -1.255 V	-1.475 V -1.400 V -1.510 V	02	2.5 2.5 2.5	--- --- ---	V
			T_C	V_{ITH}	V_{ITL}				
Low level threshold output voltage	V_{OTL}	V_{EE} = -5.2 V V_{CC} = 0 V Load = 100Ω to -2 V	+25°C +125°C -55°C	+1.8 +1.8 +2.0	+1.10 +0.80 +1.10	01	--- --- ---	-1.600 -1.525 -1.635	V
		V_{EE} = -5.2 V V_{CC} = +5.0 V Load = -12 mA	+25°C +125°C -55°C	-1.105 V -1.000 V -1.255 V	-1.475 V -1.400 V -1.510 V	02	--- --- ---	0.5 0.5 0.5	V
Low level output voltage inputs floating	V_{OLS}	V_{EE} = -5.2 V V_{CC} = +5.0 V				02	---	0.5	V
Bias voltage	V_{BB}		+25°C +125°C -55°C			02	-1.35 -1.24 -1.44	-1.23 -1.12 -1.32	V

See footnotes at end of table

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TABLE I. Electrical performance characteristics - Continued.

(Limits are valid provided circuit is in a test socket and transverse air flow of 500 linear ft/min is maintained.)

Test	Symbol	Conditions 1/ 2/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Units
Input clamp voltage	V_{IC}	V_{EE} = -5.2 V, V_{CC} = +5.0 V	01	-1.5	---	V
High level input current	I_{IH1}	V_{CC} = +5.0 V, V_{EE} = -5.2 V	01	---	50	µA
			02		196	
High level input current	I_{IH2}	V_{CC} = +5.0 V, V_{EE} = -5.2 V	02	---	200	µA
Input leakage current	I_{CBO}	V_{EE} = -5.2 V, V_{CC} = +5.0 V	02	-1.5	---	µA
Short circuit current	I_{OS}	V_{EE} = -5.2 V, V_{CC} = +5.0 V; T_C = +25°C	02	-40	-100	mA
		T_C = +125°C		-35	-100	
		T_C = -55°C		-40	-100	
Low level input current	I_{IL1}	V_{CC} = +5.0 V, V_{EE} = -5.2 V; All other inputs	01	-3.2	---	mA
	I_{IL2}	V_{CC} = +5.0 V, V_{EE} = -5.2 V; B input 3/	01	-12.8	---	mA
Low level positive supply current	I_{CCL}	V_{CC} = +5.0 V, V_{EE} = -5.2 V	01	---	25	mA
			02	---	39	
High level positive supply	I_{CCH}	V_{EE} = -5.2 V, V_{CC} = +5.0 V	01	---	18	mA
			02	---	57	
Power supply draw current	I_{EE}	V_{EE} = -5.2 V, V_{CC} = +5.0 V	01	-73	---	mA
			02	-44	---	
Input breakdown voltage	BV_{IN}	V_{EE} = -5.2 V, V_{CC} = +5.0 V; I_{IN} = 1 mA	01	5.5	---	V
Transition time, low-to-high level	t_{TLH}	V_{EEL} = -3.2 V, V_{CCH} = +7.0 V, V_{GG} = +2.0 V, Load = 100Ω to GND	01	1.0	5.0	ns
		V_{EE} = -5.2 V, V_{CC} = +5.0 V, Load = 450Ω 25 pF	02	---	5.3	ns
Transition time, high-to-low level	t_{THL}	V_{EEL} = -3.2 V, V_{CCH} = +7.0 V, V_{GG} = +2.0 V, Load = 100Ω to GND	01	1.0	5.0	ns
		V_{EE} = -5.2 V, V_{CC} = +5.0 V, Load = 450Ω 25 pF	02	---	5.3	ns

See footnotes at end of table

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TABLE I. Electrical performance characteristics - Continued.

(Limits are valid provided circuit is in a test socket and transverse air flow of 500 linear ft/min is maintained.)

Test	Symbol	Conditions 1/ 2/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Units
Propagation delay time, low-to-high level	t_{PLH}	V_{EEL} = -3.2 V V_{CCH} = +7.0 V V_{GG} = +2.0 V Load = 100Ω to GND	01	1.0	8.0	ns
		V_{EE} = -5.2 V V_{CC} = +5.0 V Load = 450Ω 25 pF	02	1.0	7.0	ns
Propagation delay time, high-to-low level	t_{PHL}	V_{EEL} = -3.2 V V_{CCH} = +7.0 V V_{GG} = +2.0 V Load = 100Ω to GND	01	1.0	8.0	ns
		V_{EE} = -5.2 V V_{CC} = +5.0 V Load = 450Ω 25 pF	02	1.0	7.0	ns

1/ Complete terminal conditions shall be as specified in table III.
2/ For voltages, the minimum and maximum limits are based on algebraic values. For currents, the minimum and maximum limits are determined by absolute value where the sign is only an indication of conventional current direction (i.e., current into terminal is positive).
3/ Applicable to "B" inputs only.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*, 2, 3, 9 10, 11	1*, 2, 3, 9
Group A test requirements (method 5005)	1, 2, 3, 9 10, 11	1, 2, 3, 9
Group B test requirements (method 5005) subgroup 5	1, 2, 3, 9 10, 11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Additional electrical subgroups for group C periodic inspections	None	10, 11
Group D end-point Electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.3c).

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3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 30 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1, through 4.4.4).

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent.

(1) Test condition D or E using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

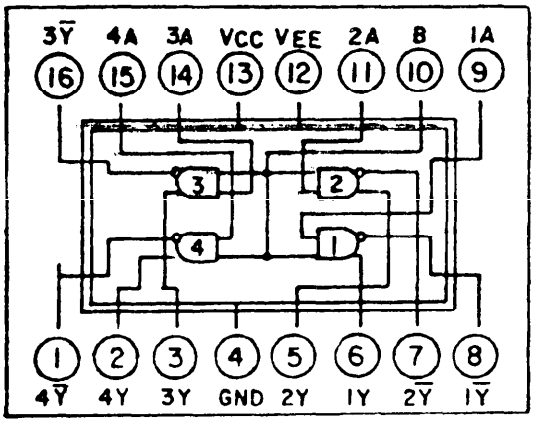
(3) Test duration: 1,000 hours, except as permitted in method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

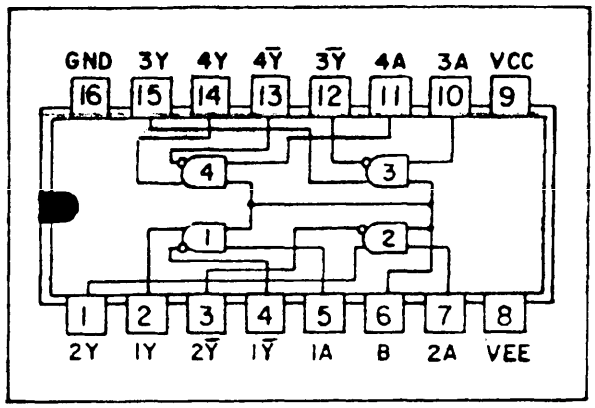
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Device type 01

Case F

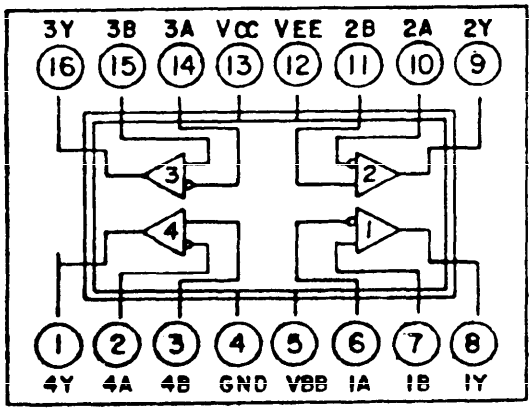


Case E



Device type 02

Case F



Case E

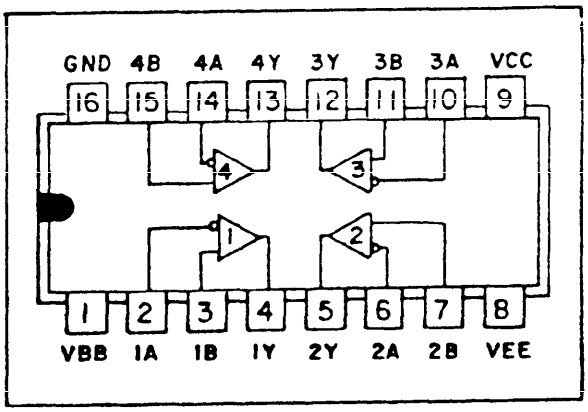
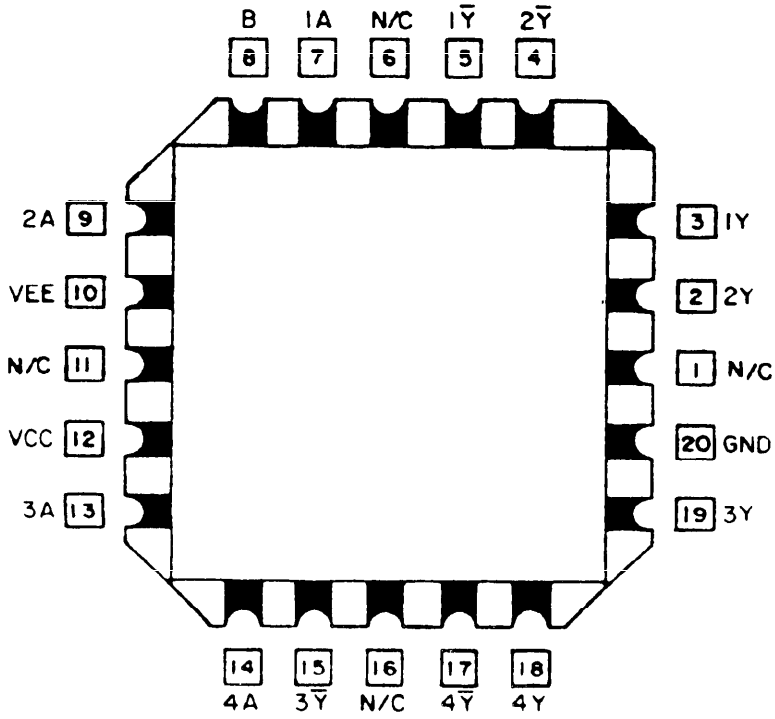


FIGURE 1. Logic diagrams and terminal connections.

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Device type 01

Case 2



Device type 02

Case 2

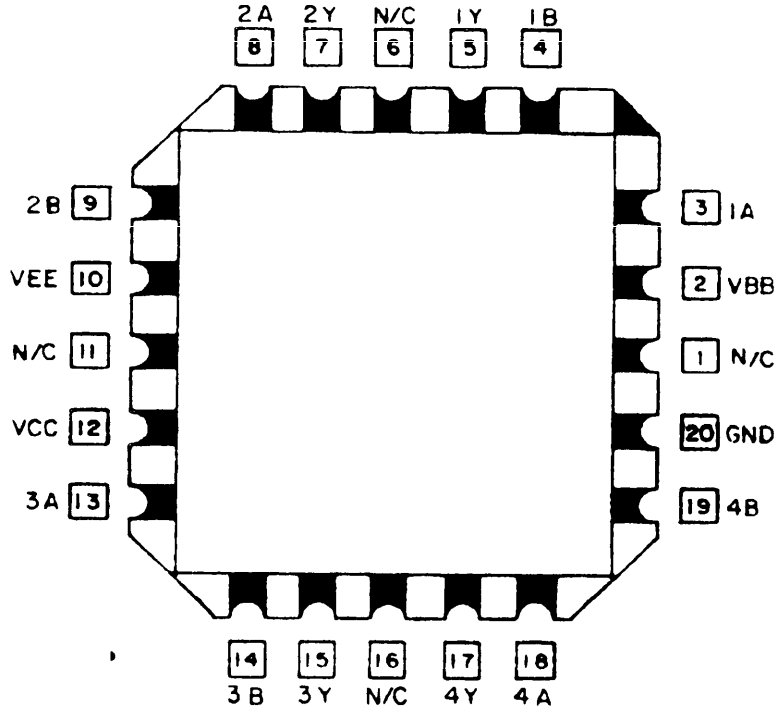


FIGURE 1. Terminal connections (top view).

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Device type 01

Truth table each gate			
Input		Output	
A	B	Y	$\overline{Y}$
L	L	L	H
H	L	L	H
L	H	L	H
H	H	H	L

Device type 02

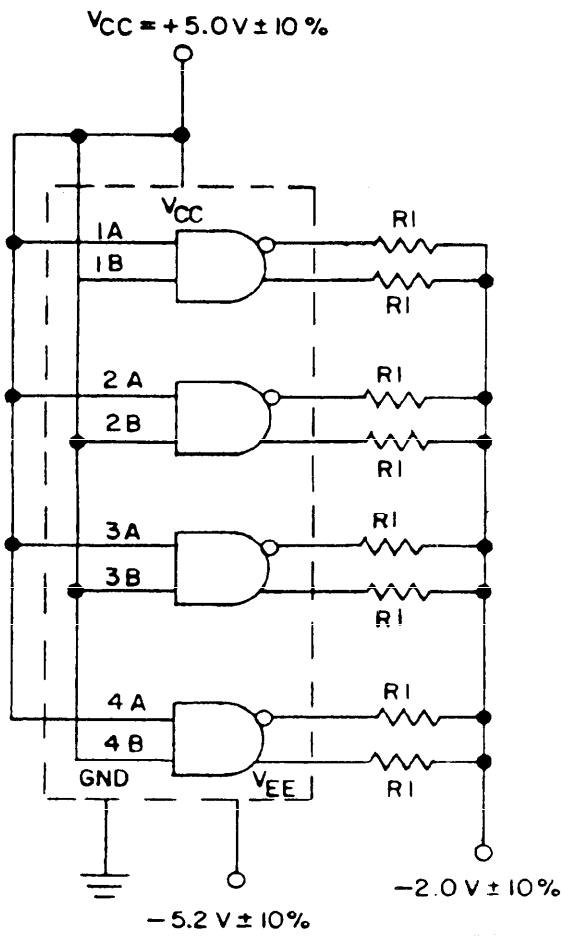
TRUTH TABLE		
Input		Output
A	B	Y
L	L	*
L	H	H
H	L	L
H	V_{BB}	L
L	V_{BB}	H
Open	Open	L
V_{BB}	H	H
V_{BB}	L	L
H	H	*

*Not defined

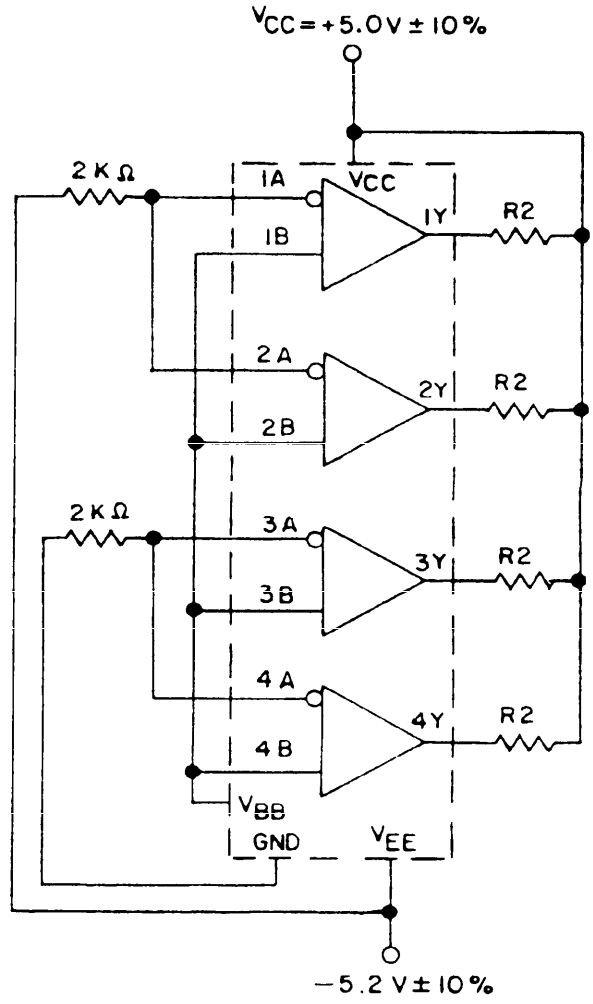
FIGURE 2. Truth tables.

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Device type 01



Device type 02



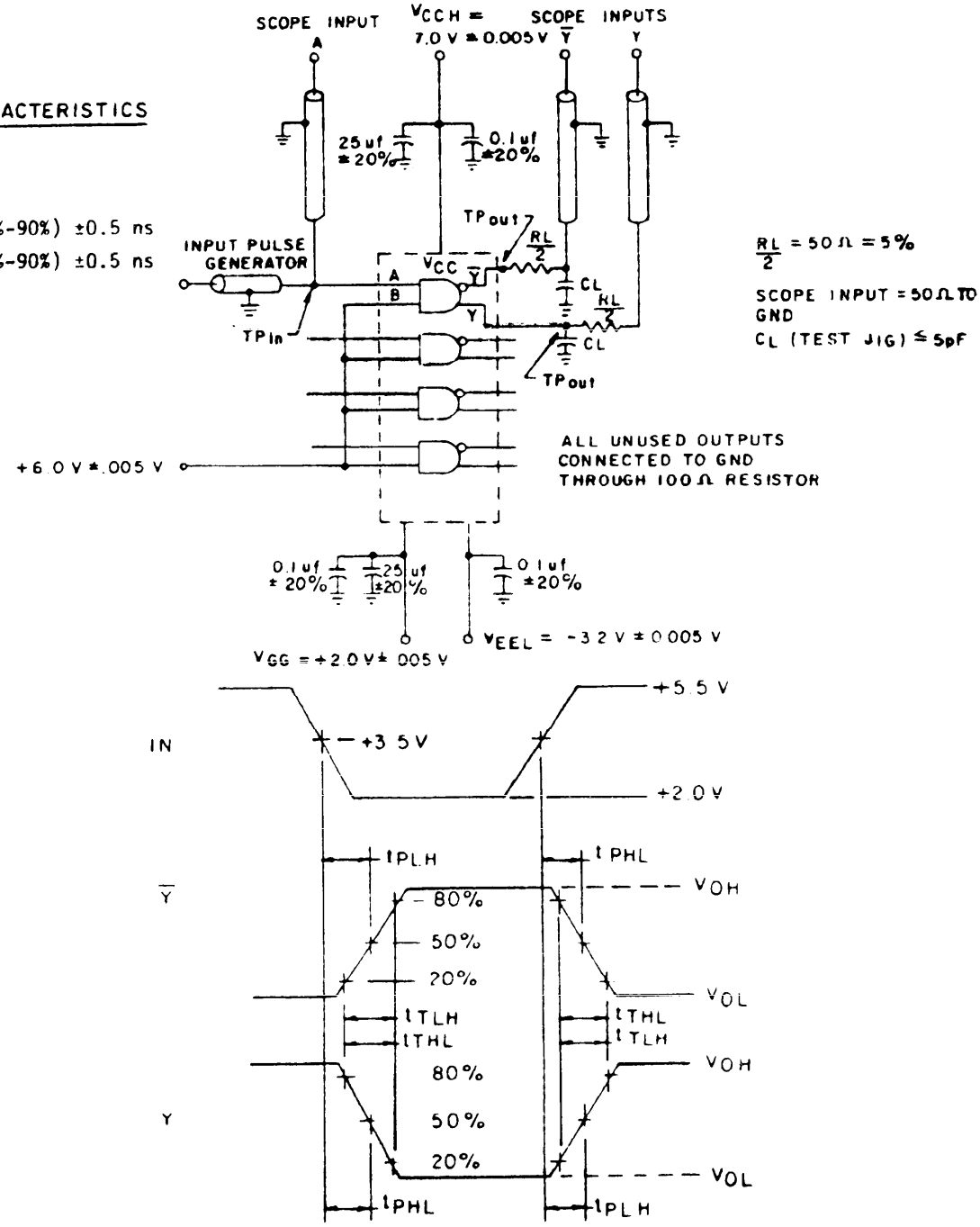
NOTES:
1. R_L = 50Ω ±5% and R_2 = 360Ω ±5%.
2. Resistors are rated at 88 mW at 125°C and 150 mW at 150°C.

FIGURE 3. Burn-in and life test circuits.

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GENERATOR CHARACTERISTICS

$Z_{out} \approx 50\Omega$
t_p = 40 ns ±2 ns
t_{THL} = 5.5 ns (10%-90%) ±0.5 ns
t_{TLH} = 5.5 ns (10%-90%) ±0.5 ns



NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50 ohm coaxial cables. Wire length should be ≤ .250 inch (6.35 mm) from t_p in to input pin and TP out to output pin.
3. Outputs not under test connected to a 100 ohm resistor to ground.

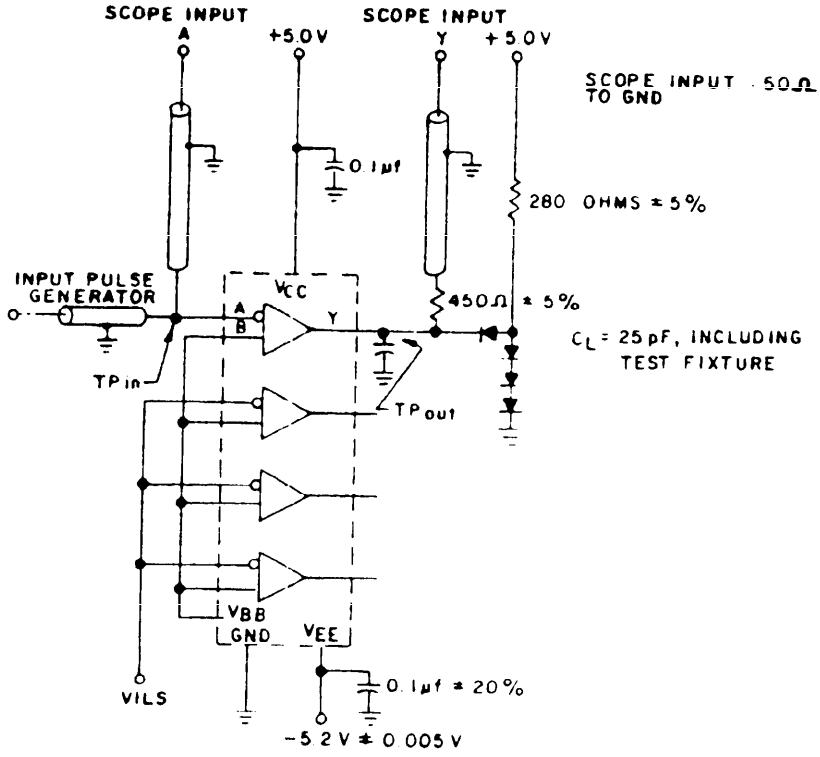
FIGURE 4. Switching test circuit for device type 01.

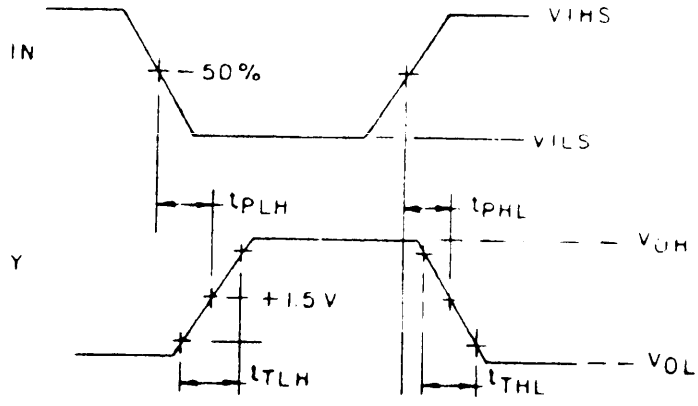
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GENERATOR CHARACTERISTICS

T_C	VIHS	VILS
25°C	-0.890 V	-1.690 V
125°C		
-55°C		

$Z_{out} \approx 50\Omega$
t_{THL} = 2.0 ns (20%-80%) ±0.2 ns
t_{TLH} = 2.0 ns (20%-80%) ±0.2 ns
t_p = 40 ns ±1.0 ns

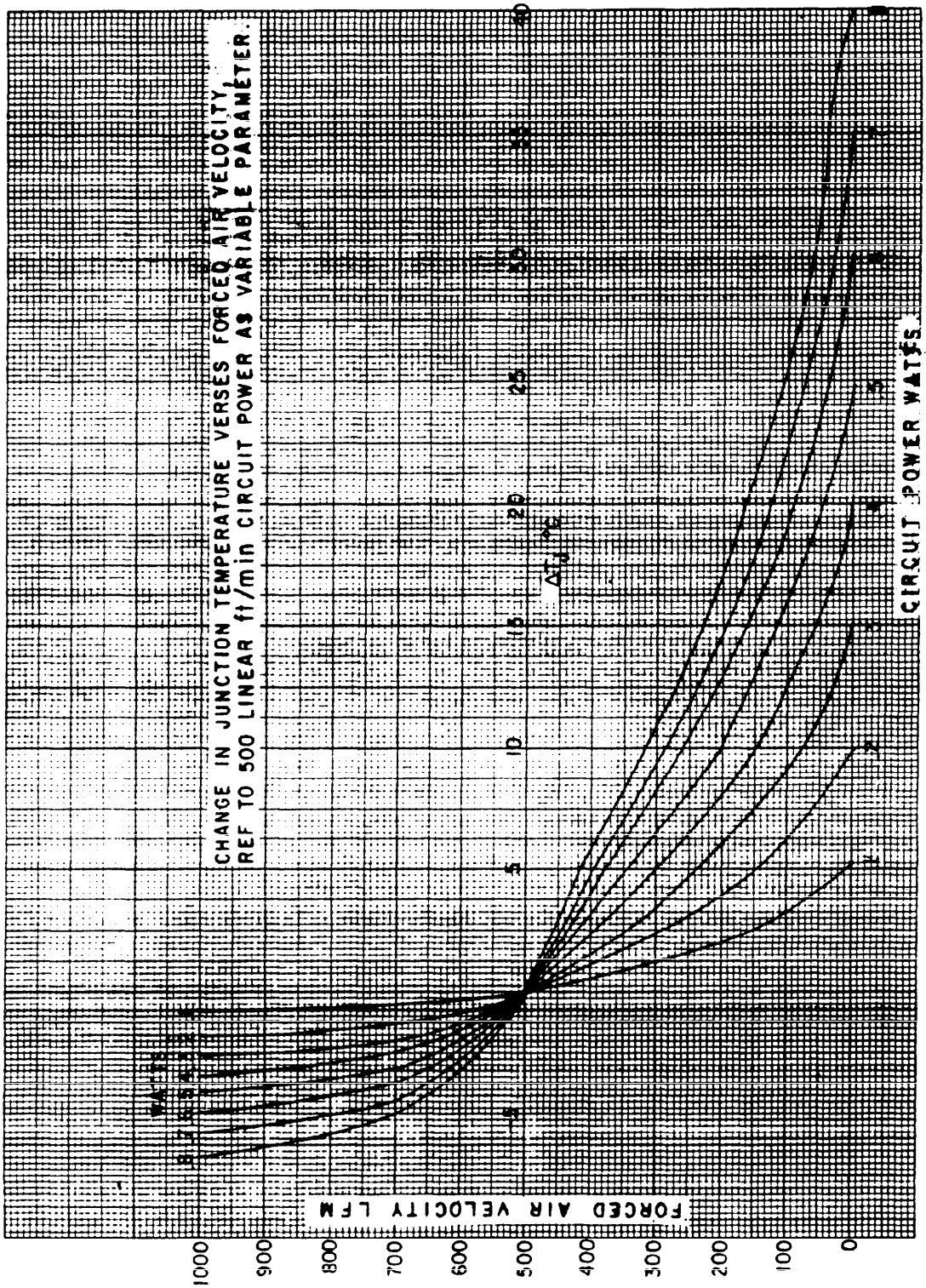




NOTES:
1. Perform test in accordance with test table; each output is tested separately.
2. All input and output cables are equal lengths of 50 ohm coaxial cables. Wire length should be ≤ .250 inch (6.35 mm) from t_p in to input pin and TP out to output pin.
3. All diodes are 1N3064 or equivalent.

FIGURE 5. Switching test circuit for device type 02.

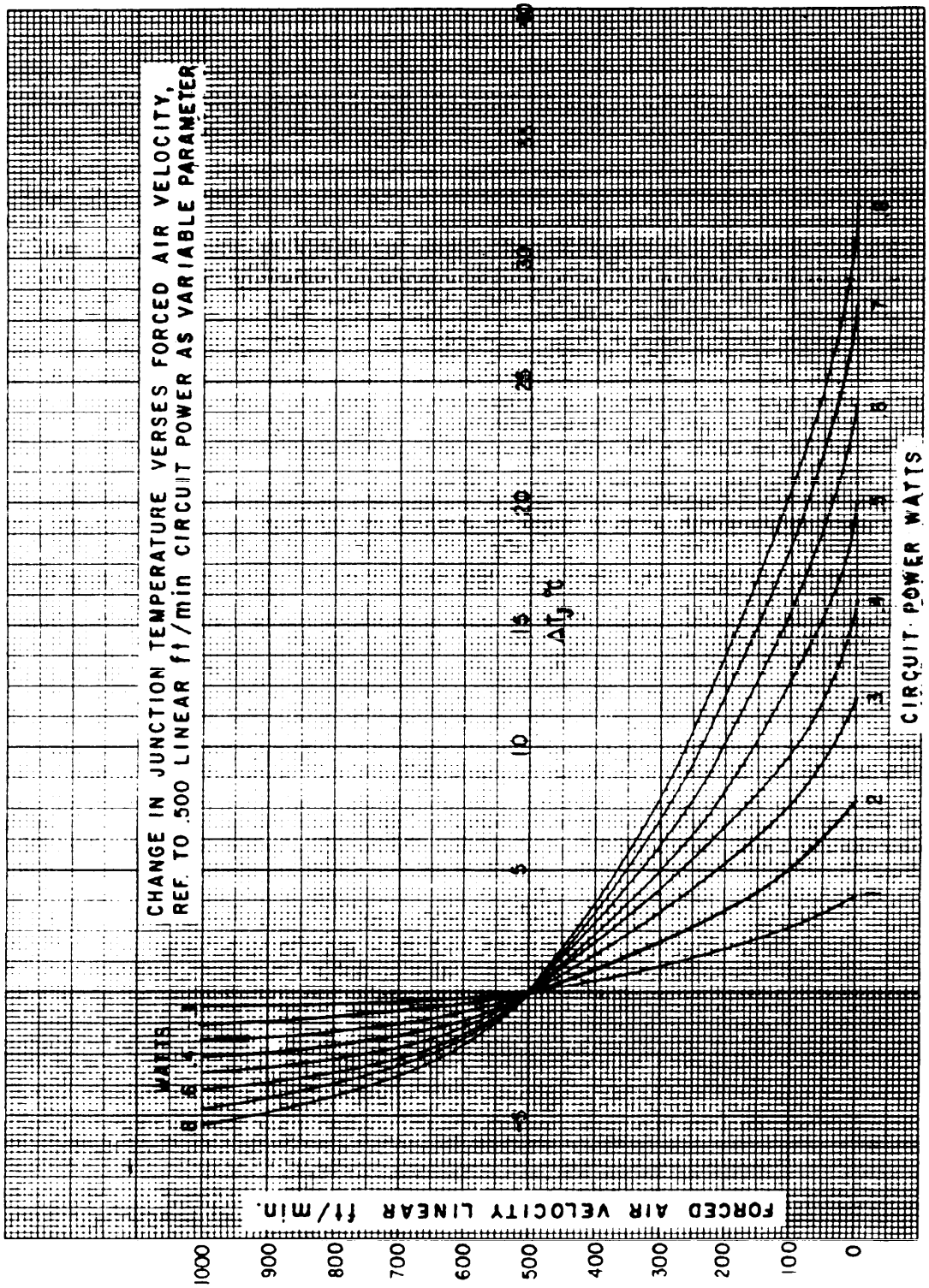
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(Case E)

FIGURE 6. Junction temperature -vs- forced air velocity.

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(Case F)

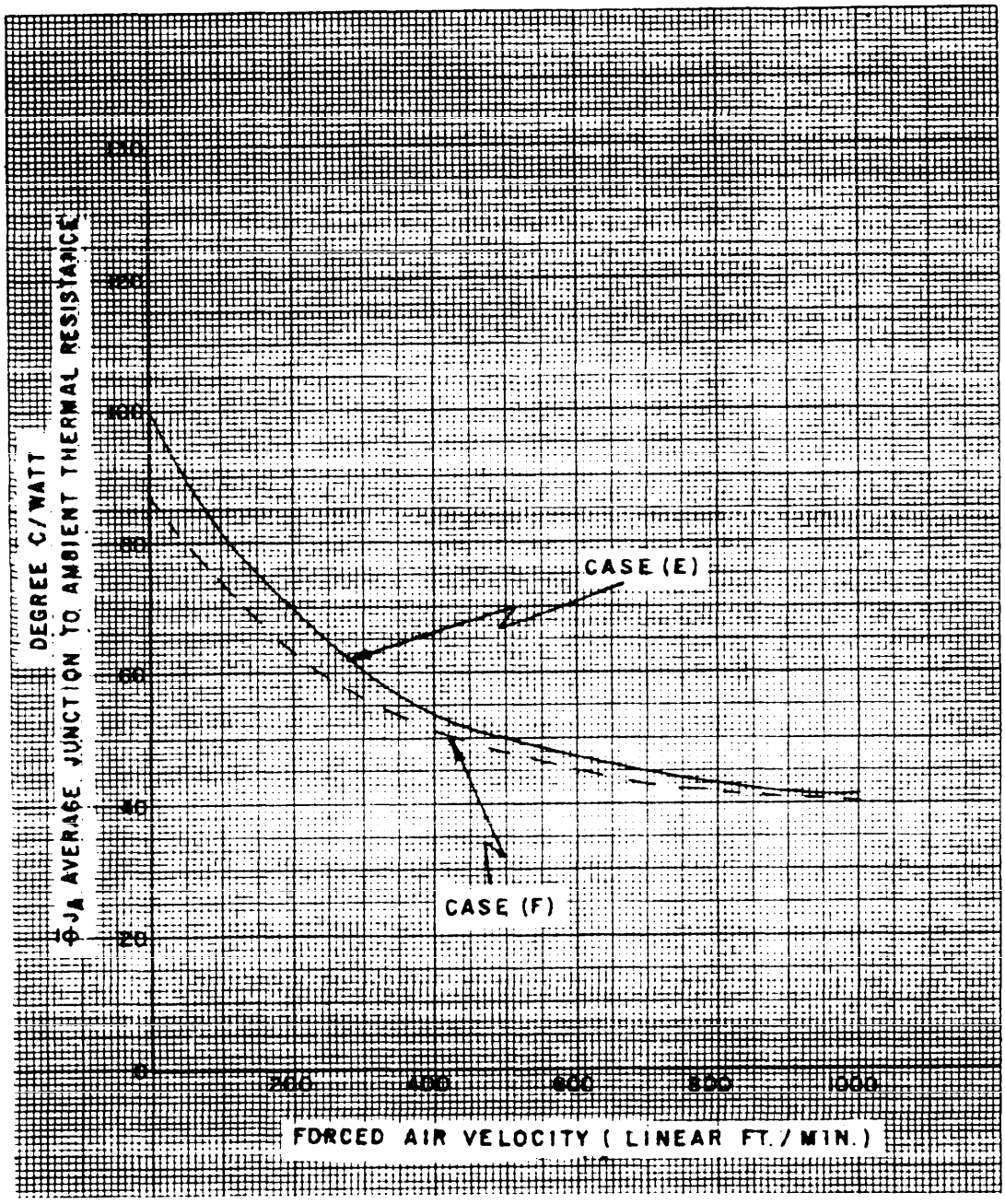
FIGURE 7. Junction temperature -vs- forced air velocity.

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	-55°C (mV/°C)		+25°C (mV/°C)		+125°C (mV/°C)	
PARAMETER	$+\Delta T_J$	$-\Delta T_J$	$+\Delta T_J$	$-\Delta T_J$	$+\Delta T_J$	$-\Delta T_J$
V_{OH} max, V_{IH_1}	1.25	1.25	1.50	1.25	1.50	1.50
V_{OH} min, V_{OTH}	1.88	1.88	1.05	1.88	1.05	1.05
V_{OL} max, V_{OTL}	0.44	0.44	0.75	0.44	0.75	0.75
V_{OL} min, V_{IL}	0.88	0.88	0.30	0.88	0.30	0.30
V_{ITH}	1.88	1.88	1.05	1.88	1.05	1.05
V_{ITL}	0.44	0.44	0.75	0.44	0.75	0.75

FIGURE 8. Adjustment coefficients for forcing function and test limit compensation.

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NOTE: ($\bar{\Theta}J_A$ -vs- forced air velocity) for case (E) and (F)
$T_J = T_C + \bar{\Theta}J_A \times P_D$ (max).

FIGURE 9. Air velocity versus average thermal resistance.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases E / Y / Z	For terminal conditions see table IV																	Measured terminal	Limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16			Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
			5	6	7	8	9	10	11	12	13	14	15	16	1	2	3	4									
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20									
		Test no.	2Y	1Y	2$\overline{Y}$	1$\overline{Y}$	1A	B	2A	V_{EE}	V_{CC}	3A	4A	3$\overline{Y}$	4$\overline{Y}$	4Y	3Y	GND		Min	Max	Min	Max	Min	Max	
V_{OH}	3006	1	LD_1	LD_1	LD_1	LD_1	V_{IH}	V_{IH}	V_{IH}	V_{EE}	V_{CC}	V_{IH}	V_{IH}	LD_1	LD_1	LD_1	LD_1	GND	1Y	-.93	-.780	-.825	-.63	-1.08	-.880	V
	"	2	"	"	"	"		"		"	"	"	"	"	"	"	"	"	2Y	"	"	"	"	"	"	"
	"	3	"	"	"	"		"		"	"	"	"	"	"	"	"	"	3Y	"	"	"	"	"	"	"
	"	4	"	"	"	"		"		"	"	"	"	"	"	"	"	"	4Y	"	"	"	"	"	"	"
	"	5	"	"	"	"		V_{IL}		"	"	"	"	"	"	"	"	"	1$\overline{Y}$	"	"	"	"	"	"	"
	"	6	"	"	"	"		"		"	"	"	"	"	"	"	"	"	2$\overline{Y}$	"	"	"	"	"	"	"
	"	7	"	"	"	"		"		"	"	"	"	"	"	"	"	"	3$\overline{Y}$	"	"	"	"	"	"	"
	"	8	"	"	"	"		"		"	"	"	"	"	"	"	"	"	4$\overline{Y}$	"	"	"	"	"	"	"
V_{OL}	3007	9	"	"	"	"				"	"	"	"	"	"	"	"	"	1Y	-1.85	-1.62	-1.82	-1.545	-1.92	-1.655	"
	"	10	"	"	"	"				"	"	"	"	"	"	"	"	"	2Y	"	"	"	"	"	"	"
	"	11	"	"	"	"				"	"	"	"	"	"	"	"	"	3Y	"	"	"	"	"	"	"
	"	12	"	"	"	"				"	"	"	"	"	"	"	"	"	4Y	"	"	"	"	"	"	"
	"	13	"	"	"	"				"	"	"	"	"	"	"	"	"	1$\overline{Y}$	"	"	"	"	"	"	"
	"	14	"	"	"	"				"	"	"	"	"	"	"	"	"	2$\overline{Y}$	"	"	"	"	"	"	"
	"	15	"	"	"	"				"	"	"	"	"	"	"	"	"	3$\overline{Y}$	"	"	"	"	"	"	"
	"	16	"	"	"	"				"	"	"	"	"	"	"	"	"	4$\overline{Y}$	"	"	"	"	"	"	"
V_{OTH}		17	"	"	"	"	V_{ITH}	V_{IH}	V_{IH}	"	"	"	"	"	"	"	"	"	1Y	-.95		-.845		-1.1		"
		18	"	"	"	"	V_{IH}	"	V_{ITH}	"	"	"	"	"	"	"	"	"	2Y	"		"		"		"
		19	"	"	"	"	"	"	V_{IH}	"	"	V_{ITH}	"	"	"	"	"	"	3Y	"		"		"		"
		20	"	"	"	"	"	"	"	"	"	V_{IH}	V_{ITH}	"	"	"	"	"	4Y	"		"		"		"
		21	"	"	"	"	"	V_{ITH}	"	"	"			"	"	"	"	"	1Y	"		"		"		"
		22	"	"	"	"	"	"	"	"	"			"	"	"	"	"	2Y	"		"		"		"
		23	"	"	"	"	"	"	"	"	"			"	"	"	"	"	3Y	"		"		"		"
		24	"	"	"	"	"	"	"	"	"			"	"	"	"	"	4Y	"		"		"		"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E, F, 2	For terminal conditions see table IV: 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits: Subgroup 1 T_C = +25°C Min	Subgroup 1 T_C = +25°C Max	Subgroup 2 T_C = +125°C Min	Subgroup 2 T_C = +125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Units
		5, 6, 7	5	6	7	8	9	10	11	12	13	14	15	16	1	2	3	4								
		2, 3, 4	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	2Y	1Y	2∇	1∇	1A	B	2A	V_{EE}	V_{CC}	3A	4A	3∇	4∇	4Y	3Y	GND								
V_{OTH}		25	LO_1	LO_1	LO_1	LO_1	V_{IH}	V_{ITL}	V_{IH}	V_{EE}	V_{CC}			LO_1	LO_1	LO_1	LO_1	GND	1∇	-.95		-.845		-1.1		V
		26	"	"	"	"	"	"	"	"	"			"	"	"	"	"	2∇	"		"		"		"
		27	"	"	"	"	"	"	"	"	"			"	"	"	"	"	3∇	"		"		"		"
		28	"	"	"	"	"	"	"	"	"			"	"	"	"	"	4∇	"		"		"		"
		29	"	"	"	"	V_{ITL}	V_{IH}	"	"	"			"	"	"	"	"	1∇	"		"		"		"
		30	"	"	"	"	V_{IH}	"	V_{ITL}	"	"			"	"	"	"	"	2∇	"		"		"		"
		31	"	"	"	"	"	"	V_{IH}	"	"	V_{ITL}		"	"	"	"	"	3∇	"		"		"		"
		32	"	"	"	"	"	"	"	"	"	V_{IH}	V_{ITL}	"	"	"	"	"	4∇	"		"		"		"
V_{OTL}		33	"	"	"	"	V_{ITL}	"	"	"	"	"	V_{IH}	"	"	"	"	"	1Y		-1.6		-1.525		-1.635	V
		34	"	"	"	"	V_{IH}	"	V_{ITL}	"	"	"	"	"	"	"	"	"	2Y		"		"		"	"
		35	"	"	"	"	"	"	V_{IH}	"	"	V_{ITL}	"	"	"	"	"	"	3Y		"		"		"	"
		36	"	"	"	"	"	"	"	"	"	V_{IH}	V_{ITL}	"	"	"	"	"	4Y		"		"		"	"
		37	"	"	"	"	"	V_{ITL}	"	"	"	"	V_{IH}	"	"	"	"	"	1Y		"		"		"	"
		38	"	"	"	"	"	"	"	"	"	"	V_{IH}	"	"	"	"	"	2Y		"		"		"	"
		39	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	3Y		"		"		"	"
		40	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	4Y		"		"		"	"
		41	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	1∇		"		"		"	"
		42	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	2∇		"		"		"	"
		43	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	3∇		"		"		"	"
		44	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	4∇		"		"		"	"
		45	"	"	"	"	V_{ITH}	V_{IH}	"	"	"	"	"	"	"	"	"	"	1∇		"		"		"	"
		46	"	"	"	"	V_{IH}	"	V_{ITH}	"	"	"	"	"	"	"	"	"	2∇		"		"		"	"
		47	"	"	"	"	"	"	"	"	"	V_{ITH}	"	"	"	"	"	"	3∇		"		"		"	"
		48	"	"	"	"	"	"	"	"	"	V_{IH}	V_{ITH}	"	"	"	"	"	4∇		"		"		"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases E	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits						Units
		F	5	6	7	3	9	10	11	12	13	14	15	15	1	2	3	4		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		L	2	3	4	5	7	8	9	10	12	13	14	15	17	19	19	20								
		Test no.	2Y	1Y	2$\overline{Y}$	1$\overline{Y}$	1A	B	2A	V_{EE}	V_{CC}	3A	4A	3$\overline{Y}$	4$\overline{Y}$	4Y	3Y	GND		Min	Max	Min	Max	Min	Max	
V_{IC}		49					-10 mA			V_{EE}	V_{CC}							GND	1A	-1.5						V
		50							-10 mA	"	"							"	2A	-						"
		51								"	"	-10 mA						"	3A	-						"
		52								"	"		10 mA					"	4A	-						"
		53						-20 mA		"	"							"	B	-						"
I_{IH1}	3010	54					V_{IH}	GND		"	"							"	1A		50		85		85	µA
	"	55						"	V_{IH}	"	"							"	2A		"		"		"	"
	"	56						"		"	"	V_{IH}						"	3A		"		"		"	"
	"	57						"		"	"		V_{IH}					"	4A		"		"		"	"
I_{IH2}	"	58					GND	V_{IH}	GND	"	"	GND	GND					"	B		200		340		340	"
I_{IL1}	3009	59					V_{IL}	"		"	"							"	1A	3.2		-5.5		-3.2		mA
	"	60						"	V_{IL}	"	"							"	2A	"		"		"		"
	"	61						"		"	"	V_{IL}						"	3A	"		"		"		"
	"	62						"		"	"		V_{IL}					"	4A	"		"		"		"
I_{IL2}	"	63						V_{IL}		"	"							"	B	12.8		-22.0		12.8		"
BV_{IN}		64					1 mA	"		"	"							"	1A	5.5		5.5		5.5		V
		65						"	1 mA	"	"							"	2A	"		"		"		"
		66						"		"	"	1 mA						"	3A	"		"		"		"
		67						"		"	"		1 mA					"	4A	"		"		"		"
		68					V_{IL}	1 mA	V_{IL}	"	"	V_{IL}	V_{IL}					"	B	-		"		"		"
I_{CCL}	3005	69					GND	GND	GND	"	"	GND	GND					"	V_{CC}		25		28		28	mA
I_{CCH}	"	70					5.0 V	5.0 V	5.0 V	"	"	5.0 V	5.0 V					"	V_{CC}		16		18		18	"
I_{EE}	"	71								"	"							"	V_{EE}	-66		-73		-73		"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases	For terminal conditions see table IV																	Measured terminal	Limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16			Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
			5	6	7	8	9	10	11	12	13	14	15	16	1	2	3	4									
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20									
		Test no.	2Y	1Y	2$\overline{Y}$	1$\overline{Y}$	1A	B	2A	V_{EE}	V_{CC}	3A	4A	3$\overline{Y}$	4$\overline{Y}$	4Y	3Y	GND		Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 Fig. 4	72	LD_2	OUT	LD_2	LD_2	IN	6.0 V		V_{EEL}	V_{CCH}			LD_2	LD_2	LD_2	LD_2	V_{GL}	1Y	1.0	6.0	1.0	8.0	1.0	8.0	ns
	"	73	LD_2	LD_2	"	OUT	IN	"		"	"			"	"	"	"	"	1$\overline{Y}$	"	"	"	"	"	"	"
	"	74	OUT	"	"	LD_2		"	IN	"	"			"	"	"	"	"	2Y	"	"	"	"	"	"	"
	"	75	LD_2	"	OUT	"		"	IN	"	"			"	"	"	"	"	2$\overline{Y}$	"	"	"	"	"	"	"
	"	76	"	"	LD_2	"		"		"	"	IN		"	"	"	OUT	"	3Y	"	"	"	"	"	"	"
	"	77	"	"	"	"		"		"	"	IN		OUT	"	"	LD_2	"	3$\overline{Y}$	"	"	"	"	"	"	"
	"	78	"	"	"	"		"		"	"		IN	LD_2	"	OUT	"	"	4Y	"	"	"	"	"	"	"
	"	79	"	"	"	"		"		"	"		IN	"	OUT	LD_2	"	"	4$\overline{Y}$	"	"	"	"	"	"	"
t_{PHL}	"	80	"	OUT	"	"	IN	"		"	"			"	LD_2	"	"	"	1Y	"	"	"	"	"	"	"
	"	81	"	LD_2	"	OUT	IN	"		"	"			"	"	"	"	"	1$\overline{Y}$	"	"	"	"	"	"	"
	"	82	OUT	"	"	LD_2		"	IN	"	"			"	"	"	"	"	2Y	"	"	"	"	"	"	"
	"	83	LD_2	"	OUT	"		"	IN	"	"			"	"	"	"	"	2$\overline{Y}$	"	"	"	"	"	"	"
	"	84	"	"	LD_2	"		"		"	"	IN		"	"	"	OUT	"	3Y	"	"	"	"	"	"	"
	"	85	"	"	"	"		"		"	"	IN		OUT	"	"	LD_2	"	3$\overline{Y}$	"	"	"	"	"	"	"
	"	86	"	"	"	"		"		"	"		IN	LD_2	"	OUT	"	"	4Y	"	"	"	"	"	"	"
	"	87	"	"	"	"		"		"	"		IN	"	OUT	LD_2	"	"	4$\overline{Y}$	"	"	"	"	"	"	"
t_{TLH}	3004 Fig. 4	88	"	OUT	"	"	IN	"		"	"			"	"	"	"	"	1Y	1.1	3.9	"	5.0	"	5.0	"
	"	89	"	LD_2	"	OUT	IN	"		"	"			"	"	"	"	"	1$\overline{Y}$	"	"	"	"	"	"	"
	"	90	OUT	"	"	LD_2		"	IN	"	"			"	"	"	"	"	2Y	"	"	"	"	"	"	"
	"	91	LD_2	"	OUT	"		"	IN	"	"			"	"	"	"	"	2$\overline{Y}$	"	"	"	"	"	"	"
	"	92	"	"	LD_2	"		"		"	"	IN		"	"	"	OUT	"	3Y	"	"	"	"	"	"	"
	"	93	"	"	"	"		"		"	"	IN		OUT	"	"	LD_2	"	3$\overline{Y}$	"	"	"	"	"	"	"
	"	94	"	"	"	"		"		"	"		IN	LD_2	"	OUT	"	"	4Y	"	"	"	"	"	"	"
	"	95	"	"	"	"		"		"	"		IN	LD_2	"	LD_2	"	"	4$\overline{Y}$	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases	Test no.	2Y	1Y	2Y	1Y	1A	6.0 V	2A	V_{EE}	V_{CC}	3A	4A	3Y	4Y	3Y	GND	Measured terminal	Limits				Units
																				Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C	
																				Min	Max	Min	Max	Min	Max	
Term	3004 Fig. 4		96	LO_2	OUT	LO_2	1Y	IN	6.0 V		V_{EEL}	V_{CCH}			LO_2	LO_2	LO_2	V_{GG}	1Y	1.1	3.9	1.0	5.0	1.0	5.0	ns
			97	LO_2	LO_2	"	OUT	IN	"		"	"			"	"	"	"	1F	"	"	"	"	"	"	"
			98	OUT	"	"	LO_2		"	IN	"	"	IN		"	"	"	"	2Y	"	"	"	"	"	"	"
			99	LO_2	"	OUT	"		"	IN	"	"	IN		"	"	OUT	"	2F	"	"	"	"	"	"	"
			100	"	"	LO_2	"		"		"	"			"	"	LO_2	"	3Y	"	"	"	"	"	"	"
			101	"	"	"	"		"		"	"		IN	OUT	"	"	"	3F	"	"	"	"	"	"	"
			102	"	"	"	"		"		"	"		IN	LO_2	"	OUT	"	4Y	"	"	"	"	"	"	"
			103	"	"	"	"		"		"	"		IN	LO_2	"	LO_2	"	4F	"	"	"	"	"	"	"

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Cases E	Cases F	Cases Z	For terminal conditions see table IV																Measured terminal	Limits						Units
																						Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.			V_{BB}	1A	1B	1Y	2Y	2A	2B	V_{EE}	V_{CC}	3A	3B	3Y	4Y	4A	4B	GND		Min	Max	Min	Max	Min	Max	
V_{OH}	3006	1				V_{IL}	V_{IH}	I_{OH}		V_{IL}	V_{IH}			V_{IL}	V_{IH}			V_{IL}	V_{IH}	GND	1Y	2.5		2.5		2.5		V
		2							I_{OH}												2Y							
		3														I_{OH}					3Y							
		4															I_{OH}				4Y							
		5				V_{ILL}	V_{ILL}	I_{OH}		V_{ILL}	V_{IL}			V_{ILL}	V_{IL}			V_{ILL}	V_{IL}		1Y							
		6							I_{OH}												2Y							
		7														I_{OH}					3Y							
		8															I_{OH}				4Y							
		9				V_{ILH}	V_{ILH}	I_{OH}		V_{ILH}	V_{IHH}			V_{ILH}	V_{IHH}			V_{ILH}	V_{IHH}		1Y							
		10							I_{OH}												2Y							
		11														I_{OH}					3Y							
		12															I_{OH}				4Y							
		13				V_{ILL}	V_{IHL}	I_{OH}		V_{ILL}	V_{IHL}			V_{ILL}	V_{IHL}			V_{ILL}	V_{IHL}		1Y							
		14							I_{OH}												2Y							
		15														I_{OH}					3Y							
		16															I_{OH}				4Y							
V_{OL}	3007	17				V_{IH}	V_{IL}	I_{OL}		V_{IH}	V_{IL}			V_{IH}	V_{IL}			V_{IH}	V_{IL}		1Y		0.5		0.5		0.5	
		18							I_{OL}												2Y							
		19														I_{OL}					3Y							
		20															I_{OL}				4Y							
		21				V_{IHH}	V_{IH}	I_{OL}		V_{IHH}	V_{IH}			V_{IHH}	V_{IH}			V_{IHH}	V_{IH}		1Y							
		22							I_{OL}												2Y							
		23														I_{OL}					3Y							
		24															I_{OL}				4Y							

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases E	For terminal conditions see table IV																Measured terminal	Limits						Units
		F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	5	6	7	8	9	10	11	12	13	14	15	16	1	7	3	4								
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20								
		Test no.	V_{BB}	1A	1B	1Y	2Y	2A	2B	V_{EE}	V_{CC}	3A	3B	3Y	4Y	4A	4B	GND		Min	Max	Min	Max	Min	Max	
V_{OL}	3007	25		V_{IHH}	V_{ILH}	I_{OL}		V_{IHH}	V_{ILH}	V_{EE}	V_{CC}	V_{IHH}	V_{ILH}			V_{IHH}	V_{ILH}	GND	1Y		0.5		0.5		0.5	V
	"	26		"	"		I_{OL}	"	"	"	"	"	"			"	"	"	2Y		"		"		"	"
	"	27		"	"			"	"	"	"	"	"	I_{OL}		"	"	"	3Y		"		"		"	"
	"	28		"	"			"	"	"	"	"	"		I_{OL}	"	"	"	4Y		"		"		"	"
	"	29		V_{IHL}	V_{ILL}	I_{OL}		V_{IHL}	V_{ILL}	"	"	V_{IHL}	V_{ILL}			V_{IHL}	V_{ILL}	"	1Y		"		"		"	"
	"	30		"	"		I_{OL}	"	"	"	"	"	"			"	"	"	2Y		"		"		"	"
	"	31		"	"			"	"	"	"	"	"	I_{OL}		"	"	"	3Y		"		"		"	"
	"	32		"	"			"	"	"	"	"	"		I_{OL}	"	"	"	4Y		"		"		"	"
V_{OH}		33	V_{BB}	V_{ITL}	V_{BB}	I_{OH}		V_{IL}	V_{BB}	"	"	V_{IL}	V_{BB}			V_{IL}	V_{BB}	"	1Y	2.5		2.5		2.5		"
		34	"	V_{IL}	"		I_{OH}	V_{ITL}	"	"	"	V_{IL}	"			"	"	"	2Y	"		"		"		"
		35	"	"	"			V_{IL}	"	"	"	V_{ITL}	"	I_{OH}		"	"	"	3Y	"		"		"		"
		36	"	"	"			"	"	"	"	V_{IL}	"		I_{OH}	V_{ITL}	"	"	4Y	"		"		"		"
V_{OTL}		37	"	V_{ITH}	"	I_{OL}		"	"	"	"	"	"			V_{IL}	"	"	1Y		0.5		0.5		0.5	"
		38	"	V_{IL}	"		I_{OL}	V_{ITH}	"	"	"	"	"			"	"	"	2Y		"		"		"	"
		39	"	"	"			V_{IL}	"	"	"	V_{ITH}	"	I_{OL}		"	"	"	3Y		"		"		"	"
		40	"	"	"			V_{IL}	"	"	"	V_{IL}	"		I_{OL}	V_{ITH}	"	"	4Y		"		"		"	"
V_{OLS}		41				I_{OL}				"	"							"	1Y		"		"		"	"
		42					I_{OL}			"	"							"	2Y		"		"		"	"
		43								"	"			I_{OL}				"	3Y		"		"		"	"
		44								"	"				I_{OL}			"	4Y		"		"		"	"
		45				I_{OL}					"							"	1Y		"		"		"	"
		46					I_{OL}				"							"	2Y		"		"		"	"
		47									"			I_{OL}				"	3Y		"		"		"	"
		48									"				I_{OL}			"	4Y		"		"		"	"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases	For terminal conditions see table IV																	Measured terminal	Limits						Units
		1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16			Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		2	2	3	4	5	7	8	9	10	11	12	13	14	15	16	17	18	19	20							
		Test no.	V_{BB}	1A	1B	1Y	2Y	2A	2B	V_{EE}	V_{CC}	3A	3B	3Y	4Y	4A	4B	GND			Min	Max	Min	Max	Min	Max	
V_{BB}		49	V_{BB}		V_{BB}				V_{BB}	V_{EE}	V_{CC}		V_{BB}				V_{BB}	GND	V_{BB}		-1.35	-1.23	-1.24	-1.12	-1.44	-1.32	V
I_{IH1}	3010	50	"	V_{IH}	V_{BB}					"	"		"				"	"	1A			115		196		196	µA
	"	51	"					V_{IH}		"	"		"				"	"	2A			"		"		"	
	"	52	"							"	"	V_{IH}	"				"	"	3A			"		"		"	
	"	53	"							"	"		"			V_{IH}	"	"	4A			"		"		"	
	"	54	"	V_{BB}	V_{IH}			V_{BB}		"	"	V_{BB}	"			V_{BB}		"	1B			"		"		"	
	"	55	"	"				"	V_{IH}	"	"	"				"		"	2B			"		"		"	
	"	56	"	"						"	"	"	V_{IH}			"		"	3B			"		"		"	
	"	57	"	"						"	"	"				"	V_{IH}	"	4B			"		"		"	
I_{CBO}		58	"	V_{EE}	V_{BB}				V_{BB}	"	"	"	V_{BB}			"	V_{BB}	"	1A	-1.0		-1.0		-1.5		"
		59	"		"			V_{EF}	"	"	"		"				"	"	2A	"		"		"		"	
		60	"		"				"	"	"	V_{EE}	"				"	"	3A	"		"		"		"	
		61	"		"				"	"	"		"			V_{EE}	"	"	4A	"		"		"		"	
		62	"	V_{BB}	V_{EE}			V_{BB}	"	"	"	V_{BB}	"			V_{BB}		"	1B	"		"		"		"	
		63	"	"				"	V_{EE}	"	"	"						"	2B	"		"		"		"	
		64	"	"				"		"	"	"	V_{EF}					"	3B	"		"		"		"	
		65	"	"				"		"	"	"					V_{EE}	"	4B	"		"		"		"	
I_{OS}		66	"	V_{IL}	V_{BB}	GND		V_{IL}	V_{BB}	"	"	V_{IL}	V_{BB}			V_{IL}	V_{BB}	"	1Y	-40	-100	-35	-100	-40	-100	nA
		67	"	"	"		GND	"	"	"	"	"	"			"	"	"	2Y	"	"	"	"	"	"	"
		68	"	"	"			"	"	"	"	"	"			"	"	"	3Y	"	"	"	"	"	"	"
		69	"	"	"			"	"	"	"	"	"			"	"	"	4Y	"	"	"	"	"	"	"
I_{CCL}	3005	70	"	"	"			"	"	"	"	"	"			"	"	"	V_{CC}		39		39		39	"
I_{CCH}	"	71	"	V_{IH}	"			V_{IH}	"	"	"	V_{IH}	"			V_{IH}	"	"	V_{CC}		52		52		52	"
I_{EE}	"	72	"		"				"	"	"		"				"	"	V_{EE}	-40		-44		-40		"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases	For terminal conditions see table IV																			Measured terminal	Limits						Units
		C	1	2	3	4	5	E	7	8	9	10	11	12	13	14	15	16					Subgroup 9 T_C = +25°C		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C		
		F	5	6	7	8	9	10	11	12	13	14	15	16	1	2	3	4											
		2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20											
		Test no.	V_{BB}	1A	1B	1Y	2Y	2A	2B	V_{EE}	V_{CC}	3A	3B	3Y	4Y	4A	4B	GND					Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 Fig. 5	73	V_{BB}	IN	V_{BB}	OUT		V_{ILS}	V_{BB}	V_{EE}	V_{CC}	V_{ILS}	V_{BB}			V_{ILS}	V_{BB}	GND	1Y				1.0	6.0	1.0	7.0	1.0	6.5	ns
	"	74	"	V_{ILS}	"		OUT	IN	"	"	"	V_{ILS}	"			"	"	"	2Y				"	"	"	"	"	"	"
	"	75	"	"	"			V_{ILS}	"	"	"	IN	"	OUT		"	"	"	3Y				"	"	"	"	"	"	"
	"	76	"	"	"			"	"	"	"	V_{ILS}	"		OUT	IN	"	"	4Y				"	"	"	"	"	"	"
t_{PHL}	"	77	"	IN	"	OUT		"	"	"	"	"	"			V_{ILS}	"	"	1Y				"	"	"	"	"	"	"
	"	78	"	V_{ILS}	"		OUT	IN	"	"	"	"	"			"	"	"	2Y				"	"	"	"	"	"	"
	"	79	"	"	"			V_{ILS}	"	"	"	IN	"	OUT		"	"	"	3Y				"	"	"	"	"	"	"
	"	80	"	"	"			"	"	"	"	V_{ILS}	"		OUT	IN	"	"	4Y				"	"	"	"	"	"	"
t_{TLH}	3004 Fig. 5	81	"	IN	"	OUT			"	"	"	"	"			V_{ILS}	"	"	1Y					3.3		5.3		4.5	"
	"	82	"	V_{ILS}	"		OUT	IN	"	"	"	"	"			"	"	"	2Y					"		"		"	"
	"	83	"	"	"			V_{ILS}	"	"	"	IN	"	OUT		"	"	"	3Y					"		"		"	"
	"	84	"	"	"			"	"	"	"	V_{ILS}	"		OUT	IN	"	"	4Y					"		"		"	"
t_{THL}	"	85	"	IN	"	OUT		"	"	"	"	"	"			V_{ILS}	"	"	1Y					"		"		"	"
	"	86	"	V_{ILS}	"		OUT	IN	"	"	"	"	"			"	"	"	2Y					"		"		"	"
	"	87	"	"	"			V_{ILS}	"	"	"	IN	"	OUT		"	"	"	3Y					"		"		"	"
	"	88	"	"	"			V_{ILS}	"	"	"	V_{ILS}	"		OUT	IN	"	"	4Y					"		"		"	"

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TABLE IV. Test Conditions for group A inspection for device type 01.

Symbol	V_{IH} (V)	V_{IL} (V)	V_{ITL} (V)	V_{ITH} (V)	V_{EE} (V)	V_{EEL} (V)	V_{CC} (V)	LD_1 LD_1	LD_2 LD_2	V_{CCH} (V)	V_{GG} (V)
T_C = +25°C	+2.4	+0.4	+1.10	+1.80	-5.2	-3.2	+5.0	100Ω to -2V	100Ω to GND	+7.0	+2.0
T_C = +125°C	+2.4	+0.4	+0.80	+1.80	-5.2	-3.2	+5.0	100Ω to -2V	100Ω to GND	+7.0	+2.0
T_C = -55°C	+2.4	+0.4	+1.10	+2.00	-5.2	-3.2	+5.0	100Ω to -2V	100Ω to GND	+7.0	+2.0

TABLE IVA. Test conditions for group A inspection for device type 02.

Symbol	V_{IH} (V)	V_{IL} (V)	V_{ITL} (V)	V_{ITH} (V)	V_{EE} (V)	V_{BB}	V_{CC} (V)	V_{CB} (V)	V_{IHH} (V)	V_{ILH} (V)	V_{IHL} (V)	V_{ILL} (V)	V_{ILS} (V)	I_{OH} (mA)	I_{OL} (mA)
T_C = +25°C	-0.780	-1.850	-1.475	-1.105	-5.2	Tie to V_{BB}	5.0	-5.2	+0.220	-0.850	-1.780	-2.850	-1.690	-2.0	+12.0
T_C = +125°C	-0.630	-1.820	-1.400	-1.000	-5.2	Tie to V_{BB}	5.0	-5.2	+0.370	-0.820	-1.630	-2.820		-2.0	+12.0
T_C = -55°C	-0.880	-1.920	-1.510	-1.255	-5.2	Tie to V_{BB}	5.0	-5.2	+0.120	-0.920	-1.880	-2.920		-2.0	+12.0

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

BV_{IN}	Input breakdown voltage
GND	Ground zero voltage potential.
V_{OTH}	High-level threshold output voltage.
V_{OTL}	Low-level threshold output voltage.
V_{ITH}	High-level threshold input voltage.
V_{ITL}	Low-level threshold input voltage.
V_{EEL}	Shifted power supply voltage for the purpose of ac testing.
V_{OLS}	Input protection tests.
I_{CBO}	Input leakage.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	10524
02	10525

6.6 Test limit compensation examples.

a. A device which has a power dissipation of 100 mW in case F is to be tested under a zero air flow condition. On figure 7 ΔT_J between 500 linear ft/min and zero air flow is +4°C. In order to adjust the various parameter limits, use figure 8 which defines the limit adjustment coefficients for ΔT_J. To adjust V_{OH}(max) at -55°C, use the $+\Delta T_J$ column of the -55°C portion of figure 8 and locate the coefficient corresponding to V_{OH}(max). This value is 1.25 mV/°C. Multiply at the ΔT_J by the coefficient and algebraically add it to the -55°C V_{OH}(max) limit from table III.

$$V_{OH}(max)\ (adjusted\ limit) = (+4^{\circ}C \times 1.25\ mV/^{\circ}C) + (-880\ mV)$$
$$= 5\ mV - 880\ mV = -875\ mV$$
$$Use\ -875\ mV$$

Follow the same procedure to adjust the remaining parameters at -55°C as well as all parameters at +25°C and +125°C.

b. A device with a power dissipation of 150 mW in case E is to be tested at an air flow of 200 linear ft/min and the 25°C testing is to be accomplished at a case temperature of +20°C. On figure 6 ΔT_J due to air flow is +3°C. The ΔT_J due to ambient temperature change is -5°C (25-20). Therefore the total ΔT_J = -5 +3 = -2°C. Using figure 8, find the +25°C, $-\Delta T_J$ column. To adjust V_{OL} (max) locate the limit coefficient corresponding to V_{OL} (max) for a negative ΔT_J, this value is 0.44 mV°C. Multiply the ΔT_J by the coefficient and algebraically add it to the +25°C V_{OL} (max) limit from table III.

$$V_{OL}\ (max)\ (adjusted\ limit) = (-2^{\circ}C) \times (0.44\ mV/^{\circ}C) + (-1620\ mV)$$
$$= -.88\ mV - 1620\ mV = -1620.88\ mV$$
$$Use\ -1621\ mV$$

Follow the same procedure to adjust the remaining parameters at +25°C.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/63A	MICROCIRCUITS, DIGITAL, ECL, QUAD TRANSLATOR, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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